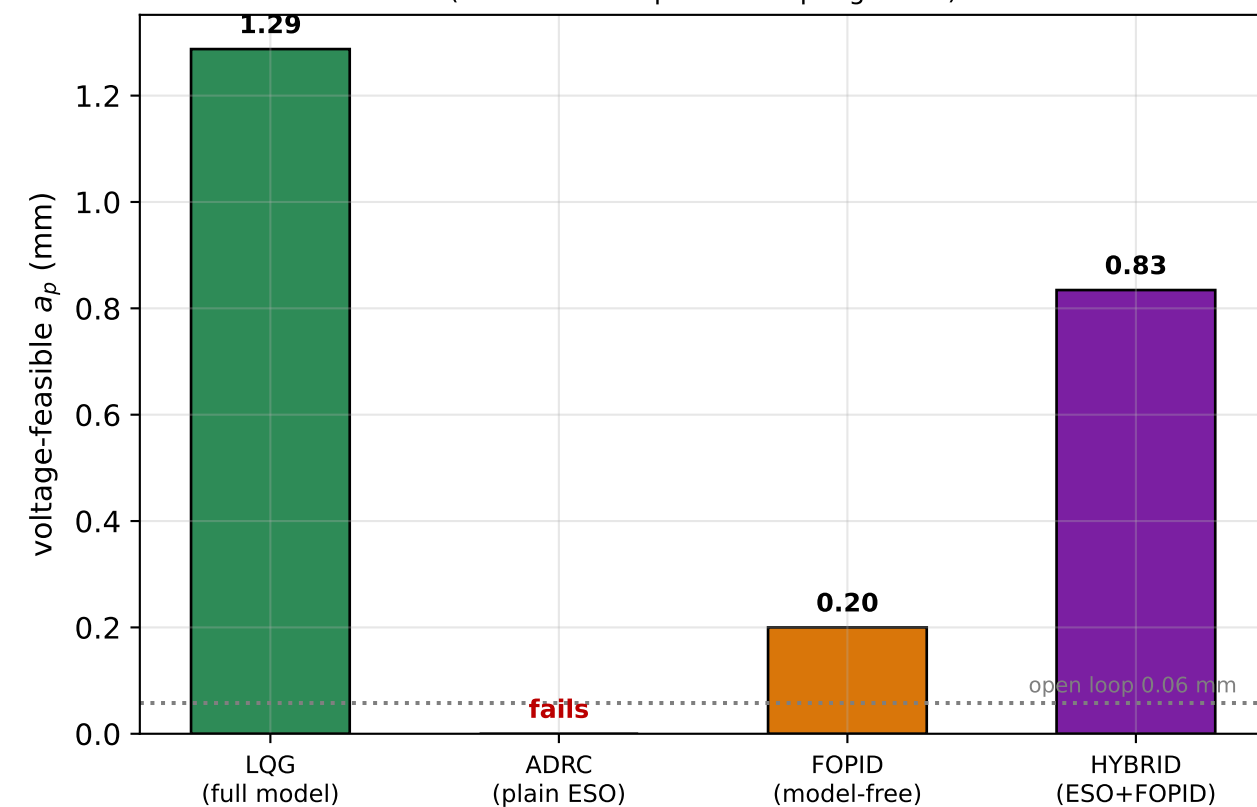
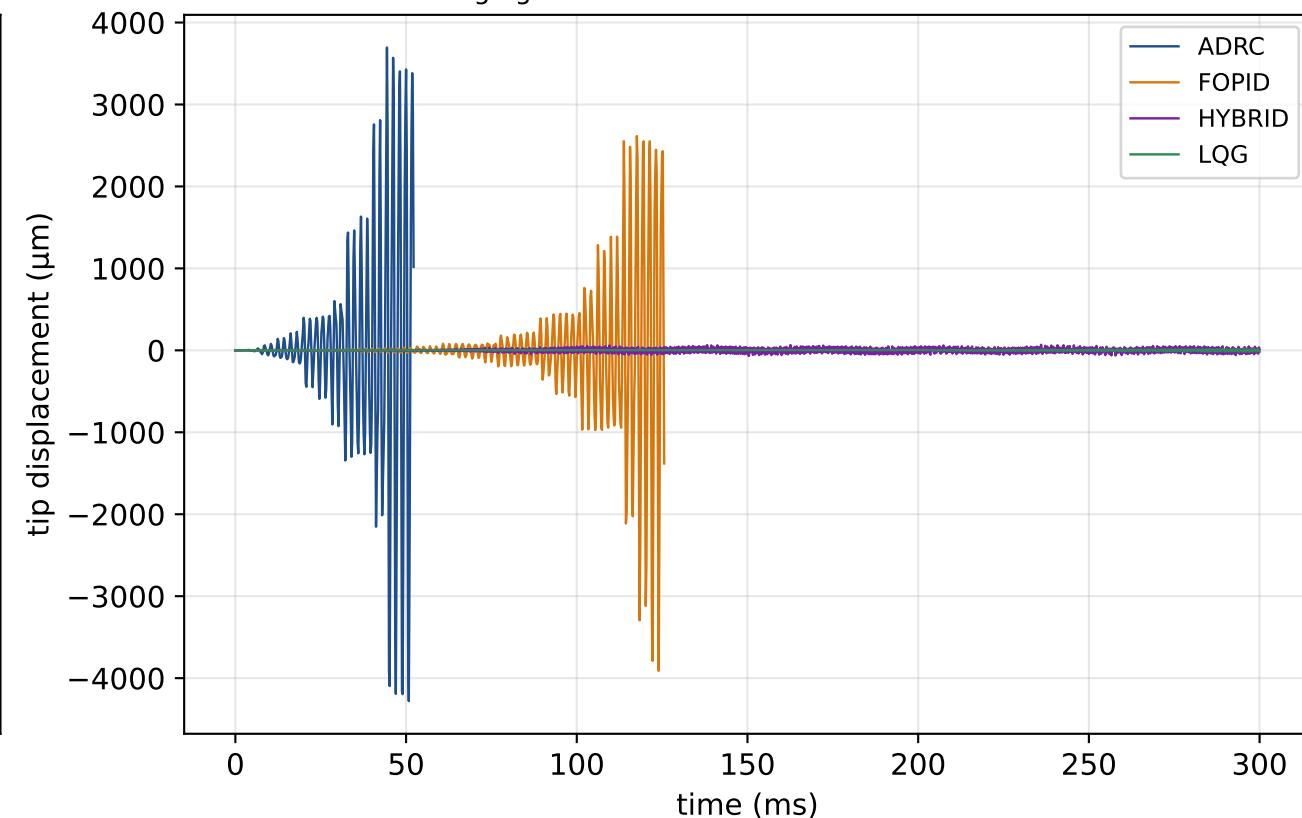


Tip-sensor-only comparison (the article's measurement): LQG / ADRC / FOPID / HYBRID (band-limited ESO + FOPID)

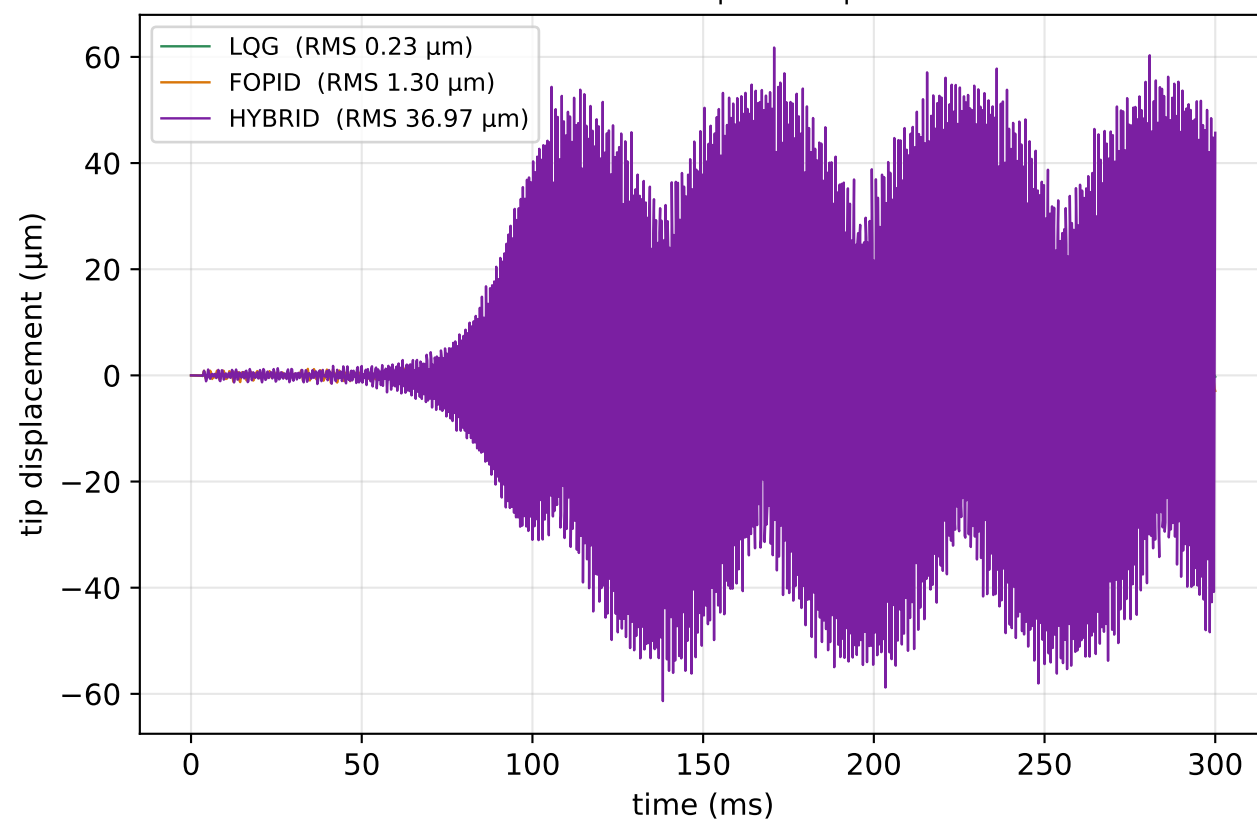
(a) feasible depth, all controllers on the article's tip sensor
(non-minimum phase: coupling -+--+)



(b) tip response at $a_p = 0.5$ mm
diverging traces terminate at the chatter threshold



(c) rejection at $a_p = 0.15$ mm (all stable)
does the ESO branch improve on plain FOPID?



(d) robustness: feasible depth under frequency drift
(tip sensor; plain ADRC infeasible everywhere)

